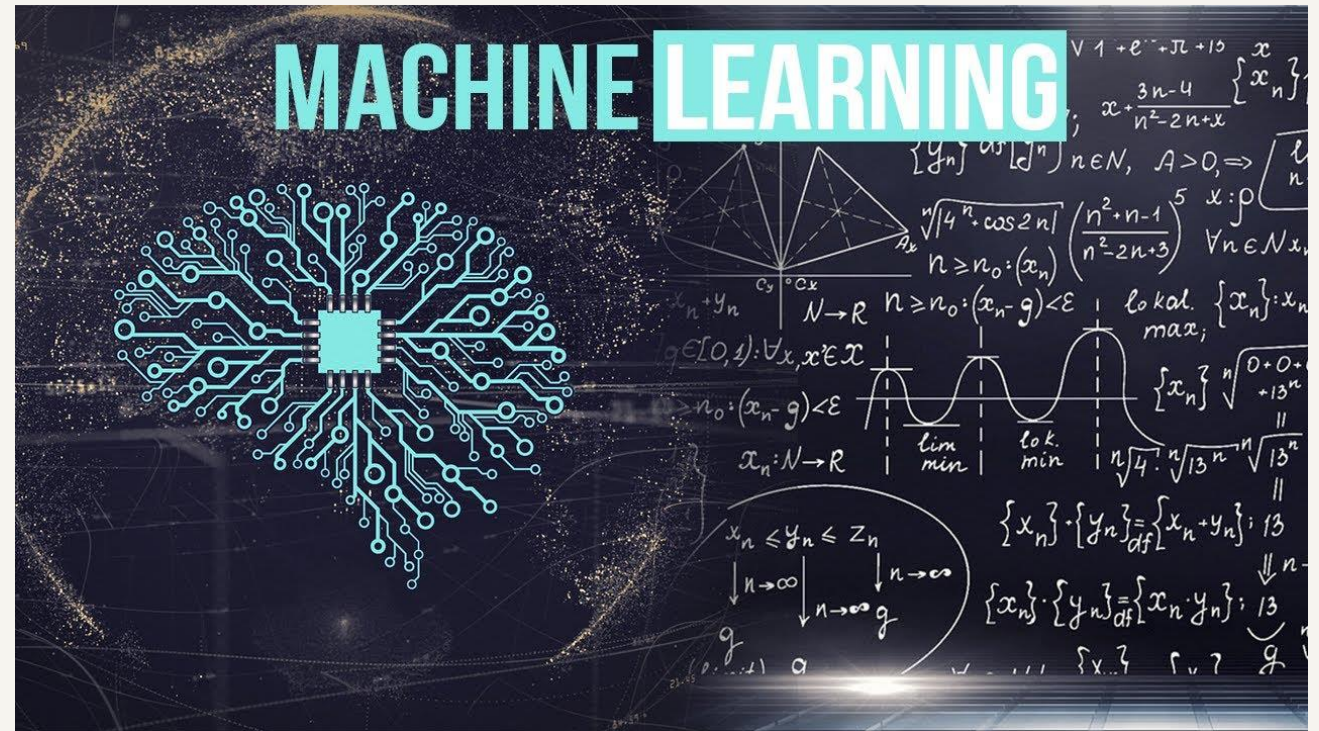


Sampling & Population



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- What is Statistics?
- What is Sampling?
- What is Population?
- What is Data and its Type?
- Qualitative Data
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What is Statistics?

Important Terms in Statistics

- In statistics, we generally want to study a **population**. You can think of a **population** as a collection of persons, things, or objects under study. To study the population, we select a **sample**.
- The idea of sampling is to select a portion (or subset) of the larger population and study that portion (the sample) to gain information about the population. Data are the result of sampling from a population.

What is Statistics? - Example

- If you wished to compute the overall grade point average at your school, it would make sense to select a sample of students who attend the school. The data collected from the sample would be the students' grade point averages.

Population: All the students at your school

Sample: a sample of 50 students

- In presidential elections, opinion poll samples of 1,000–2,000 people are taken. The opinion poll is supposed to represent the views of the people in the entire country.

Population: All the people in the entire country

Sample: Sample of 1000-2000 people

What is Statistics? - Example

- City of Houston wants to know if the annual household income in the city is higher than national average. The statisticians collect data from 1500 families.

Population: All the households in Houston

Sample: 1500 families who are being sampled.

- An automobile manufacturer wanted to know if more than 50% of the US drivers own at least a domestic car. This company surveyed 10,000 drivers over USA.

Population: All the drivers in the US

Sample: 10,000 drivers who were being surveyed

What is Statistics?

- From the sample data, we can calculate a statistic. A statistic is a number that represents a property of the sample.
- For example,
 - if we consider one math class to be a sample of the population of all math classes, then the average number of points earned by students in that one math class at the end of the term is an example of a statistic.
 - The statistic is an estimate of a population parameter.
 - A parameter is a number that is a property of the population.
 - Since we considered all math classes to be the population, then the average number of points earned per student over all the math classes is an example of a parameter.

Population: all math classes

Sample: One of the math classes

Parameter: Average number of points earned per student over all math classes

What is Statistics?

- One of the main concerns in the field of statistics is how accurately a statistic estimates a parameter.
- The accuracy really depends on **how well the sample represents the population**.
- The sample must contain the characteristics of the population in order to be a **representative sample**.
- We are interested in both the sample statistic and the population parameter in inferential statistics.
- (In a later chapters/semesters, we will use the sample statistic to tests the validity of the established population parameter.)

What is Statistics?

- A variable, notated by capital letters such as X and Y , is a characteristic of interest for each person or thing in a population.
- Variables may be **numerical** or **categorical**.
- Numerical variables take on values with equal units such as weight in pounds and time in hours.
- Categorical variables place the person or thing into a category.

What is Statistics?

What is statistics?

Statistics Definition: (Science of Average and their Estimate) Statistics is the science of collecting, organizing, presenting, analyzing and interpreting data for specific purpose to help in making more effective decision.

Why study statistics:

To make more effective decision for the betterment of individual, society, business, nature and so on

Statistical Analysis:

Statistical analysis is implemented to manipulate, summarize, and investigate data, so that useful decision-making information results are obtained.

Two type of statistics:

- 1.Descriptive Statistics (used to describe the basic features of the data)
- 2.Inferential statistics (aims at learning characteristics of the population from a sample)

What is Data and its type?

Data is a set of collected or recorded facts of particular subject.

Data in general terms refer to facts and statistics collected together for reference or analysis.

Types of Data:

1. **Qualitative Data**
2. **Quantitative Data**

Qualitative Data

1. Qualitative Data:

“Data Associated with the quality in different categories”. Data is measurements, each fall into one of several categories. (Hair Color, ethnic groups and other attributes of the population)

(a). Nominal Data: “With no inherent order or ranking”

~ Data with no inherent order or ranking such as gender or race, such kind of data called Nominal Data.

(b). Ordinal Data: “with an order series”

Nominal data are categorical in nature, while ordinal data are in between categorical and quantitative.

This is because we sometimes assign quantitative values to ordinal data.

Although we cannot perform any arithmetic operation with ordinal numbers, it is quite different from nominal data which does not have any quantitative value at all

Ex: For example, very hot, hot, cold, very cold, warm are all nominal data when considered individually. But when placed on a scale and arranged in a given order (very hot, hot, warm, cold, very cold), they are regarded as ordinal data.

Qualitative Data

•Examples

Examples of nominal data include country, gender, race, hair color etc. of a group of people, while that of ordinal data includes having a position in class as “First” or “Second”. Note that the nominal data examples are nouns, with no order to them while ordinal data examples come with a level of order.

Consider the two examples below:

1.How was your customer service experience?

2. How was your customer service experience?

- Good
- Neutral
- Bad

The data to be collected from Example I. is a nominal data, while that of II. is an ordinal data.

Qualitative Data

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The data to be collected from Example I. is a nominal data, while that of II. is an ordinal data.

Qualitative Data

E.g. How old are you?

This is an example of a nominal data collection that takes a numerical value as an input. However, we cannot perform any arithmetic operation on this input.

E.g.2. Rate your customer service experience on a scale of 1-5 (Lowest-Highest)

- 1
- 2
- 3
- 4
- 5

Qualitative Data

QUE:

Is Age Nominal or Ordinal Data?

- Age can be both nominal and ordinal data depending on the question types. I.e "How old are you" is used to collect nominal data while "Are you the firstborn or What position are you in your family" is used to collect ordinal data.
- Age becomes ordinal data when there's some sort of order to it.

Quantitative Data

2. Quantitative Data:

“Data associated with Quantity which can be measured” ~ Data measured on a numeric scale (distance travelled to college, the number of children in a family etc.)

(a). Discrete Data: “Based on count, finite number of values possible and value cannot be subdivided”

~ Data which can be categorized into classification, data which is based upon counts, there are only a finite number of values possible, and values cannot be subdivided meaningfully, such kind of data is called Discrete Data.

(b). # Continuous Data: “measured on a continuum or a scale, value which can be subdivided into finer increments”

~ Data which can be measured on a continuum or a scale, data which can be have almost any numeric value and can be subdivided into finer and finer increments, such kind of data is called Continuous Data.

Sampling Techniques

Sampling Techniques

1. Probability Sampling
2. Non-Probability Sampling

Sampling Techniques

Sampling Techniques

1. Probability Sampling :

This Sampling technique uses randomization to make sure that every element of the population gets an equal chance to be part of the selected sample. It's alternatively known as random sampling.

(a) Random Sampling :

Every element has an equal chance of getting selected to be the part sample. It is used when we don't have any kind of prior information about the target population.

For example: Random selection of 20 students from class of 50 student. Each student has equal chance of getting selected. Here probability of selection is $1/50$.

Sampling Techniques

(b) Systematic Sampling

Here the selection of elements is systematic and not random except the first element. Elements of a sample are chosen at regular intervals of population. All the elements are put together in a sequence first where each element has the equal chance of being selected.

For a sample of size n , we divide our population of size N into subgroups of k elements.

We select our first element randomly from the first subgroup of k elements.

To select other elements of sample, perform following:

We know number of elements in each group is k i.e N/n

So if our first element is n_1 then

Second element is n_1+k i.e n_2

Third element n_2+k i.e n_3 and so on..

Taking an example of $N=20$, $n=5$

No of elements in each of the subgroups is N/n i.e $20/5 = 4 = k$

Now, randomly select first element from the first subgroup.

If we select $n_1 = 3$

$n_2 = n_1 + k = 3 + 4 = 7$

$n_3 = n_2 + k = 7 + 4 = 11$

Sampling Techniques

Sampling Techniques

(c) Stratified Sampling

This technique divides the elements of the population into small subgroups (strata) based on the similarity in such a way that the elements within the group are homogeneous and heterogeneous among the other subgroups formed. And then the elements are randomly selected from each of these strata. We need to have prior information about the population to create subgroups.

Sampling Techniques

2. Non-Probability Sampling :

It does not rely on randomization. This technique is more reliant on the researcher's ability to select elements for a sample. Outcome of sampling might be biased and makes difficult for all the elements of population to be part of the sample equally. This type of sampling is also known as non-random sampling.

(a) Snowball Sampling:

This technique is used in the situations where the population is completely unknown and rare. Therefore we will take the help from the first element which we select for the population and ask him to recommend other elements who will fit the description of the sample needed.

So, this referral technique goes on, increasing the size of population like a snowball.

For example:

It's used in situations of highly sensitive topics like HIV Aids where people will not openly discuss and participate in surveys to share information about HIV Aids.

Not all the victims will respond to the questions asked so researchers can contact people they know or volunteers to get in touch with the victims and collect information

Helps in situations where we do not have the access to sufficient people with the characteristics we are seeking. It starts with finding people to study.

Sampling Techniques

(b) Quota Sampling:

This type of sampling depends on some pre-set standard. It selects the representative sample from the population. Proportion of characteristics/ trait in sample should be same as population. Elements are selected until exact proportions of certain types of data are obtained or sufficient data in different categories is collected.

For example:

If our population has 45% females and 55% males then our sample should reflect the same percentage of males and females.

(c) Judgement sampling

This is based on the intention or the purpose of study. Only those elements will be selected from the population which suits the best for the purpose of our study.

For Example:

If we want to understand the thought process of the people who are interested in pursuing master's degree then the selection criteria would be "Are you interested for Masters in..?"

All the people who respond with a "No" will be excluded from our sample.

Sampling Techniques

(d) Convenience Sampling

Here the samples are selected based on the availability. This method is used when the availability of sample is rare and also costly. So based on the convenience samples are selected.

For example:

Researchers prefer this during the initial stages of survey research, as it's quick and easy to deliver results.

Chapter 2: Sampling and Population



Thank You
Vishal Modi